Name
Date
Course

More Complicated Logarithmic Equations

Exponent Rules Learned in Pre-Algebra

I) $x^a \cdot x^b = x^{a+b}$ Example: $x^2 \cdot x^5 = x^{2+5} = x^7$

II) $\frac{x^a}{x^b} = x^{a-b}$, $x \neq 0$ Example: $\frac{x^9}{x^3} = x^{9-3} = x^6$

New Exponent Rules

III) $(xy)^a = x^a \cdot y^a$ Example: $(2m)^4 = 2^4 \cdot m^4$

IV) $\left(\frac{x}{y}\right)^a = \frac{x^a}{y^a}$, $y \neq 0$ Example: $\left(\frac{n}{3}\right)^7 = \frac{n^7}{3^7}$

V) $(x^a)^b = x^{ab}$ Example: $(x^{10})^3 = x^{10 \cdot 3} = x^{30}$

Strange New Exponent Rules

VI) $x^1 = x$ Example: $3^1 = 3$

VII) $x^0 = 1$, $x \neq 0$ Example: $5^0 = 1$

VIII) $x^{-a} = \frac{1}{x^a}$, $x \neq 0$ Example: $x^{-2} = \frac{1}{x^2}$

IX) $x^{\frac{1}{a}} = \sqrt[a]{x}$ Example: $x^{\frac{1}{3}} = \sqrt[3]{x}$

X) $x^{\frac{b}{a}} = (\sqrt[a]{x})^b$ or $\sqrt[a]{x^b}$ Example: $x^{\frac{5}{7}} = (\sqrt[7]{x})^5$ or $\sqrt[7]{x^5}$

Review

Example I) Solve: $\log_2 x = 3$

$$2^3 = x$$

$$\boxed{8 = x}$$

Example II) Solve: $2 = \log_x 25$

$$x^2 = 25$$

$$\sqrt{x^2} = \pm\sqrt{25}$$

$$x = \pm 5$$

$$\boxed{x = 5} \text{ or } x \neq -5$$

Example III) Solve: $x = \log_3 27$

$$3^x = 27$$

$$3^x = 3^3$$

$$\boxed{x = 3}$$

More Complicated Logarithmic Equations

Solve. Write the Roman numeral of each exponent rule that you use.

① $\log_3(5x-6)=2$

$3^2 = 5x-6$

$9 = 5x-6$

$\boxed{3=x}$

VI

② $1=\log_4\left(\frac{x}{3}+5\right)$

$4^1 = \frac{x}{3}+5$

$4 = \frac{x}{3}+5$

$\boxed{-3=x}$

③ $\log_8(7+6x)=0$

$8^0 = 7+6x$

VII

$1 = 7+6x$

$\boxed{-1=x}$

VIII

④ $-2=\log_3\left(5-\frac{x}{4}\right)$

$3^{-2} = 5-\frac{x}{4}$

$\frac{1}{3^2} = 5-\frac{x}{4}$

$\frac{1}{9} = 5-\frac{x}{4}$

$-\frac{44}{9} = -\frac{x}{4}$

$\boxed{176/9=x}$

⑤ $\log_3(2x-1)=4$

$3^4 = 2x-1$

$81 = 2x-1$

$\boxed{41=x}$

VII

⑥ $0=\log_4\left(\frac{x}{7}+8\right)$

$4^0 = \frac{x}{7}+8$

$1 = \frac{x}{7}+8$

$\boxed{-49=x}$

⑦ $1=\log_9\left(\frac{x}{3}-2\right)$

$9^1 = \frac{x}{3}-2$

VI

$9 = \frac{x}{3}-2$

$\boxed{33=x}$

VIII

⑧ $\log_2\left(1+\frac{x}{5}\right)=-3$

$2^{-3} = 1+\frac{x}{5}$

$\frac{1}{2^3} = 1+\frac{x}{5}$

$\frac{1}{8} = 1+\frac{x}{5}$

$-\frac{7}{8} = \frac{x}{5}$

$\boxed{-35/8=x}$

(9) $-5 = -6 + \dfrac{\log_6\left(\frac{x}{3}-4\right)}{2}$

$1 = \dfrac{\log_6\left(\frac{x}{3}-4\right)}{2}$

$2 = \log_6\left(\frac{x}{3}-4\right)$

IX

$6^2 = \frac{x}{3} - 4$

$36 = \frac{x}{3} - 4$

$\boxed{120 = x}$

(10) $2\log_3(-1+4x)+7=8$

$2\log_3(-1+4x)=1$

$\log_3(-1+4x)=\frac{1}{2}$

$3^{\frac{1}{2}} = -1+4x$

$\sqrt{3} = -1+4x$

$\sqrt{3}+1 = 4x$

$\boxed{\dfrac{\sqrt{3}+1}{4} = x}$

(11) $\dfrac{\log_4(9-2x)}{4} + \dfrac{1}{8} = \dfrac{1}{4}$

$\dfrac{\log_4(9-2x)}{4} = \dfrac{1}{8}$

$\log_4(9-2x) = \frac{1}{2}$

IX

$4^{1/2} = 9-2x$

$\sqrt{4} = 9-2x$

$2 = 9-2x$

$\boxed{\dfrac{7}{2} = x}$

(12) $6 = 4 + 3\log_{27}(5x+8)$

$2 = 3\log_{27}(5x+8)$

$\frac{2}{3} = \log_{27}(5x+8)$

$27^{\frac{2}{3}} = 5x+8$

X

$\left(\sqrt[3]{27}\right)^2 = 5x+8$

$3^2 = 5x+8$

$9 = 5x+8$

$\boxed{\dfrac{1}{5} = x}$

(13) $\dfrac{\log_2\left(\frac{x}{5}+3\right)}{4} + 2 = 3$

$\dfrac{\log_2\left(\frac{x}{5}+3\right)}{4} = 1$

$\log_2\left(\frac{x}{5}+3\right) = 4$

$2^4 = \frac{x}{5}+3$

$16 = \frac{x}{5}+3$

$\boxed{65 = x}$

* Must use rule X and the distributive property.

(14) $12 = 2 + 8\log_7\left(\frac{x}{3}+6\right)$

$10 = 8\log_7\left(\frac{x}{3}+6\right)$

$\frac{10}{8} = \log_7\left(\frac{x}{3}+6\right)$

$\frac{5}{4} = \log_7\left(\frac{x}{3}+6\right)$

$7^{5/4} = \frac{x}{3}+6$

X

$\left(\sqrt[4]{7}\right)^5 = \frac{x}{3}+6$

$\left(\sqrt[4]{7}\right)^5 - 6 = \frac{x}{3}$

$\boxed{3\left(\sqrt[4]{7}\right)^5 - 18 = x}$

15) $-\frac{1}{8} = -\frac{1}{16} + \frac{\log_{81}(2-7x)}{4}$

$-\frac{1}{16} = \frac{\log_{81}(2-7x)}{4}$

$-\frac{4}{16} = \log_{81}(2-7x)$

$-\frac{1}{4} = \log_{81}(2-7x)$

$81^{-\frac{1}{4}} = 2-7x$

VIII $\frac{1}{81^{1/4}} = 2-7x$

IX $\frac{1}{\sqrt[4]{81}} = 2-7x$

$\frac{1}{3} = 2-7x$

$-\frac{5}{3} = -7x$

$\boxed{\frac{5}{21} = x}$

16) $-14\log_9(6x+2)+4 = -3$

$-14\log_9(6x+2) = -7$

$\log_9(6x+2) = \frac{1}{2}$

$9^{\frac{1}{2}} = 6x+2$

IX $\sqrt{9} = 6x+2$

$3 = 6x+2$

$\boxed{\frac{1}{6} = x}$

Using Logarithm Properties V–VII to Solve Logarithmic Equations

Logarithm Property V

$\log_k(AB) = \log_k A + \log_k B$

Logarithm Property VI

$\log_k\left(\frac{A}{B}\right) = \log_k A - \log_k B$

Logarithm Property VII

$\log_k A^B = B\log_k A$

Solve without a calculator. Write the Roman numeral of each exponent rule or logarithm property that you use.

17) $\log_4 5 + \log_4 x = 2$

L V $\log_4(5x) = 2$

$4^2 = 5x$

$16 = 5x$

$\boxed{\frac{16}{5} = x}$

18) $1 = \log_3 x - \log_3 4$

L VI $1 = \log_3\left(\frac{x}{4}\right)$

$3^1 = \frac{x}{4}$

E VI $3 = \frac{x}{4}$

$\boxed{12 = x}$

(19) $0 = \log_5 y + \log_5 4$

LV $0 = \log_5(4y)$

$5^0 = 4y$

EVII $1 = 4y$

$\boxed{\frac{1}{4} = y}$

(20) $\log_2 b - \log_2 3 = 5$

LVI $\log_2\left(\frac{b}{3}\right) = 5$

$2^5 = \frac{b}{3}$

$32 = b/3$

$\boxed{96 = b}$

(21) $\log_6 x + \log_6 3 = 2$

LV $\log_6(3x) = 2$

$6^2 = 3x$

$36 = 3x$

$\boxed{12 = x}$

(22) $3 = \log_2 y - \log_2 10$

LVI $3 = \log_2\left(\frac{y}{10}\right)$

$2^3 = \frac{y}{10}$

$8 = y/10$

$\boxed{80 = y}$

(23) $\log_4 9 + \log_4 x = 0$

LV $\log_4(9x) = 0$

$4^0 = 9x$

EVII $1 = 9x$

$\boxed{\frac{1}{9} = x}$

(24) $1 = \log_7 b - \log_7 2$

LVI $1 = \log_7\left(\frac{b}{2}\right)$

$7^1 = \frac{b}{2}$

EVI $7 = \frac{b}{2}$

$\boxed{14 = b}$

(25) $2\log_4 x + 3\log_4 2 = 3$

L VII $\log_4 x^2 + \log_4 2^3 = 3$

$\log_4 x^2 + \log_4 8 = 3$

L V $\log_4(8x^2) = 3$

$4^3 = 8x^2$

$64 = 8x^2$

$8 = x^2$

$\pm\sqrt{8} = \sqrt{x^2}$

$\pm 2\sqrt{2} = x$

$\boxed{x = 2\sqrt{2}}$ or $x \neq -2\sqrt{2}$

(26) $0 = \log_2 y - 4\log_2 3$

L VIII $0 = \log_2 y - \log_2 3^4$

$0 = \log_2 y - \log_2 81$

L VI $0 = \log_2\left(\frac{y}{81}\right)$

$2^0 = \frac{y}{81}$

E VIII $1 = y/81$

$\boxed{81 = y}$

(27) $2\log_{125} 2 + 3\log_{125} x = \frac{1}{3}$

L VII $\log_{125} 2^2 + \log_{125} x^3 = \frac{1}{3}$

$\log_{125} 4 + \log_{125} x^3 = \frac{1}{3}$

L V $\log_{125}(4x^3) = \frac{1}{3}$

$125^{\frac{1}{3}} = 4x^3$

E IX $\sqrt[3]{125} = 4x^3$

$5 = 4x^3$

$\frac{5}{4} = x^3$

$\sqrt[3]{\frac{5}{4}} = \sqrt[3]{x^3}$

$\boxed{\sqrt[3]{5/4} = x}$

(28) $1 = 2\log_5 y - 4\log_5 2$

L VIII $1 = \log_5 y^2 - \log_5 2^4$

$1 = \log_5 y^2 - \log_5 16$

L VII $1 = \log_5\left(\frac{y^2}{16}\right)$

$5^1 = \frac{y^2}{16}$

E VII $5 = \frac{y^2}{16}$

$80 = y^2$

$\pm\sqrt{80} = \sqrt{y^2}$

$\pm\sqrt{80} = y$

$\pm 4\sqrt{5} = y$

$\boxed{4\sqrt{5} = y}$ or $-4\sqrt{5} \neq y$

(29) $4\log_3 2 + \log_3 y = 0$

LVIII $\log_3 2^4 + \log_3 y = 0$

$\log_3 16 + \log_3 y = 0$

LV $\log_3(16y) = 0$

$3^0 = 16y$

EVIII $1 = 16y$

$\boxed{1/16 = y}$

(30) $\frac{1}{2} = 6\log_4 b - 5\log_4 2$

LVIII $\frac{1}{2} = \log_4 b^6 - \log_4 2^5$

$\frac{1}{2} = \log_4 b^6 - \log_4 32$

LVI $\frac{1}{2} = \log_4\left(\frac{b^6}{32}\right)$

$4^{\frac{1}{2}} = \frac{b^6}{32}$

EIX $\sqrt{4} = b^6/32$

$2 = b^6/32$

$64 = b^6$

$\pm\sqrt[6]{64} = \sqrt[6]{b^6}$

$\pm 2 = b$

$\boxed{b=2}$ or $b \neq -2$

(31) $2\log_9 x + 4\log_9 3 = 2$

LVIII $\log_9 x^2 + \log_9 3^4 = 2$

$\log_9 x^2 + \log_9 81 = 2$

LV $\log_9(81x^2) = 2$

$9^2 = 81x^2$

$81 = 81x^2$

$1 = x^2$

$\pm\sqrt{1} = \sqrt{x^2}$

$\pm 1 = x$

$\boxed{x=1}$ or $x \neq -1$

(32) $1 = 5\log_2 m - 2\log_2 5$

LVIII $1 = \log_2 m^5 - \log_2 5^2$

$1 = \log_2 m^5 - \log_2 25$

LVI $1 = \log_2\left(\frac{m^5}{25}\right)$

$2^1 = \frac{m^5}{25}$

EVI $2 = \frac{m^5}{25}$

$50 = m^5$

$\sqrt[5]{50} = \sqrt[5]{m^5}$

$\boxed{\sqrt[5]{50} = m}$

33) $2\log_2 5+\log_2 x+\log_2(3x)=3$

LVIII $\log_2 5^2+\log_2 x+\log_2(3x)=3$

$\log_2 25+\log_2 x+\log_2(3x)=3$

LVI $\log_2(25x)+\log_2(3x)=3$

LV $\log_2(25x\cdot 3x)=3$

$\log_2(75x^2)=3$

$2^3=75x^2$

$8=75x^2$

$\frac{8}{75}=x^2$

$\pm\sqrt{\frac{8}{75}}=\sqrt{x^2}$

$\pm\frac{\sqrt{8}}{\sqrt{75}}=x$

$\pm\frac{2\sqrt{2}}{5\sqrt{3}}=x$

$\pm\frac{2\sqrt{6}}{5\cdot 3}=x$

$\boxed{\frac{2\sqrt{6}}{15}=x}$

34) $2=\log_5 y^2+3\log_5 2+\log_5 3$

LVIII $2=\log_5 y^2+\log_5 2^3+\log_5 3$

$2=\log_5 y^2+\log_5 8+\log_5 3$

LVI $2=\log_5(8y^2)+\log_5 3$

LV $2=\log_5(8y^2\cdot 3)$

$2=\log_5(24y^2)$

$5^2=24y^2$

$25=24y^2$

$\frac{25}{24}=y^2$

$\pm\sqrt{\frac{25}{24}}=\sqrt{y^2}$

$\pm\frac{\sqrt{25}}{\sqrt{24}}=y$

$\pm\frac{5}{\sqrt{24}}=y$

$y=\pm\frac{5}{2\sqrt{6}}$

$y=\pm\frac{5\sqrt{6}}{2\cdot 6}$

$\boxed{y=\pm\frac{5\sqrt{6}}{12}}$

35) $1=\log_9 b^2-\log_9 4+3\log_9 2$

LVIII $1=\log_9 b^2-\log_9 4+\log_9 2^3$

$1=\log_9 b^2-\log_9 4+\log_9 8$

LVII $1=\log_9\left(\frac{b^2}{4}\right)+\log_9 8$

LV $1=\log_9\left(\frac{b^2}{4}\cdot 8\right)$

$1=\log_9\left(\frac{8b^2}{4}\right)$

$1=\log_9(2b^2)$

$9^1=2b^2$

EVI $9=2b^2$

$\frac{9}{2}=b^2$

$\pm\sqrt{\frac{9}{2}}=\sqrt{b^2}$

$\pm\frac{\sqrt{9}}{\sqrt{2}}=b$

$\pm\frac{3}{\sqrt{2}}=b$

$\boxed{\pm\frac{3\sqrt{2}}{2}=b}$

36) $\log_4 x+\log_4(3x)-4\log_4 3=0$

LVIII $\log_4 x+\log_4(3x)-\log_4 3^4=0$

$\log_4 x+\log_4(3x)-\log_4 81=0$

LV $\log_4(x\cdot 3x)-\log_4 81=0$

$\log_4(3x^2)-\log_4 81=0$

LVII $\log_4\left(\frac{3x^2}{81}\right)=0$

$4^0=3x^2/81$

EVIII $1=3x^2/81$

$81=3x^2$

$27=x^2$

$\pm\sqrt{27}=\sqrt{x^2}$

$\pm 3\sqrt{3}=x$

$\boxed{3\sqrt{3}=x}$